

A9. Assembler tilida murakkab arifmetik amallarni bajarish

Amaliy ishning maqsadi

Talabalarda assembler tilida murakkab arifmetik hisoblashlarni algoritmik fikrlash asosida bajarish, registrlardan samarali foydalanish, DX:AX jufti, signed/unsigned hisoblashlar va real amaliy masalalarni assemblerga moslashtirish ko'nikmalarini shakllantirish.

Kerakli jihozlar va vositalar

- Kompyuter yoki noutbuk
- EMU8086 / DOSBox / NASM (ixtiyoriy)
- Windows Calculator (Programmer mode)
- Qog'oz va ruchka

Nazariy qisqacha eslatma

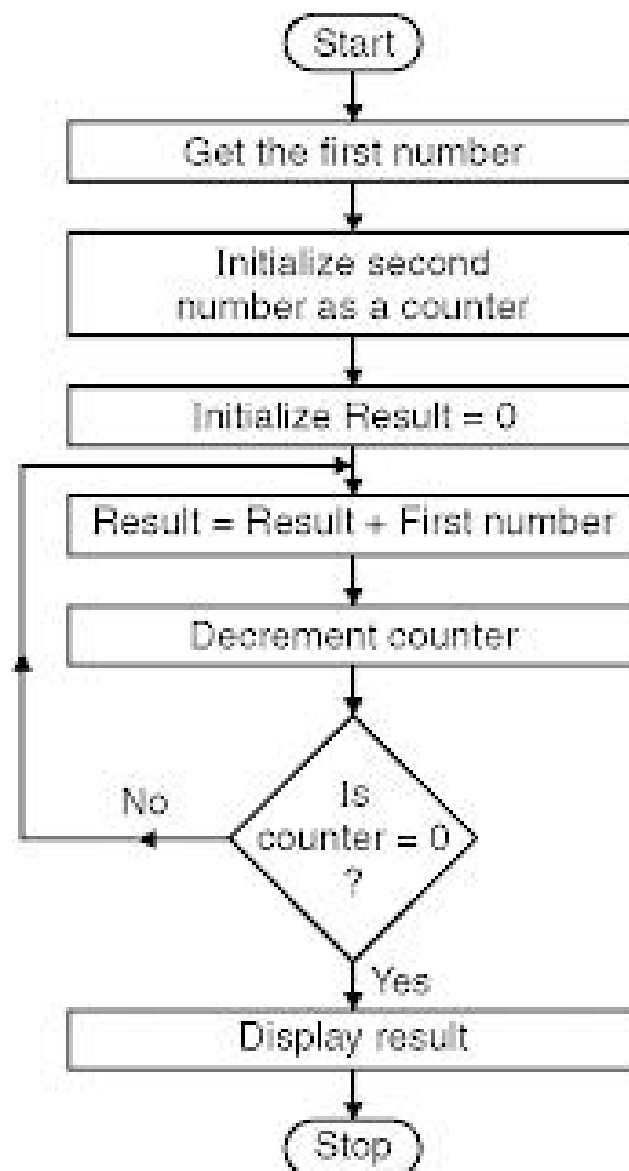
Assembler tilida murakkab arifmetik masalalar:

- **bir nechta registrlar ishtirokida**
- **ketma-ket buyruqlar orqali**
- **ALU va flaglar nazorati ostida**

bajariladi. Yuqori darajadagi tillarda bitta operator bo'lsa, assemblerda bu **butun algoritm** bo'lishi mumkin.

1-topshiriq. DX:AX juftligi bilan ishlash (chuqurlashtirish)

Quyidagi rasm asosida **DX:AX** registr juftligi vazifasini tushuning.



Topshiriq:

1. Nima sababdan 16-bit ko'paytirishda natija 32-bit bo'ladi?
2. AX va DX qaysi qismini saqlaydi?
3. Agar $DX \neq 0$ bo'lsa, bu nimani anglatadi?

2-topshiriq. Ko'p bosqichli ifodani assemblerda hisoblash

Berilgan ifoda:

$$(A \times B + C) \div D (A \times B + C) \div D$$

Algoritm:

1. $A \times B$
2. Natijaga C qo'shish

3. D ga bo'lish

Assembler mantiqi:

```
MOV AX, A
MUL B
ADD AX, C
MOV BX, D
DIV BX
```

Topshiriq:

A=6, B=4, C=12, D=3

→ Natijani **qo'l bilan** va **assembler mantiqda** hisoblang.

3-topshiriq. Belgili va belgisiz hisoblashni solishtirish

Quyidagi ikkita holatni tahlil qiling:

```
MOV AX, -8
MOV BX, 2
MUL BX
MOV AX, -8
MOV BX, 2
IMUL BX
```

Signed vs. Unsigned Multiplication

- Signed multiplication and unsigned multiplication lead to different results. Consider two bytes:

11111111 and 00000001

Signed multiplication of these two numbers gives:

11111111 (-1) * 00000001 (1) = 1111111111111111 (-1)

Unsigned multiplication of these two numbers gives:

11111111 (255) * 00000001 (1) = 0000000011111111 (255)

- REMEMBER:
 - If you multiply an 8-bit number by an 8-bit number, you will get a 16-bit number!
 - If you multiply a 16-bit number by a 16-bit number, you will get a 32-bit number!

Multiplication and Division Instructions

- MUL Instruction
- IMUL Instruction
- DIV Instruction
- Signed Integer Division (IDIV Instruction)
- Implementing Arithmetic Expressions
- Decimal Input & Output

MUL Instruction

- The MUL (unsigned multiply) instruction multiplies an 8/16 bit operand by AL/AX
- The instruction formats are:
MUL multiplier
- Multiplier can be register or memory

Implied operands:

Multiplicand	Multiplier	Product
AL	<i>r/m8</i>	AX
AX	<i>r/m16</i>	DX:AX

Savollar:

1. Natijalar nima uchun farq qiladi?
2. Qaysi holatda noto'g'ri natija chiqadi?
3. Qaysi buyruq qachon ishlatiladi?

4-topshiriq. Qoldiq orqali sonni tekshirish (amaliy masala)

Masala:

Berilgan son **juft yoki toq** ekanligini aniqlang.

Assembler mantiqi:

```
MOV AX, N
MOV BX, 2
DIV BX
; agar DX = 0 → juft
```

Topshiriq:

N = 17 va N = 24 uchun tahlil qiling.

5-topshiriq. Minimal va maksimal qiymatni aniqlash

Masala:

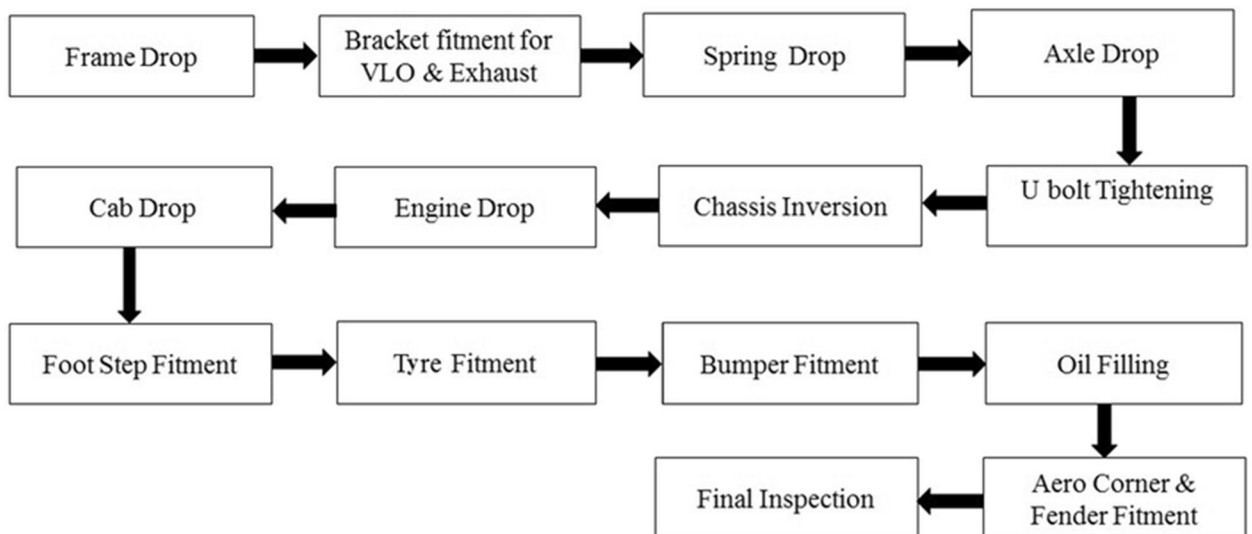
Ikki son berilgan: A va B
Kattasini aniqlang.

Assembler algoritmi:

```
MOV AX, A
CMP AX, B
JG A_KATTA
MOV AX, B
A_KATTA:
```

Instruction	Programming Equivalent	Jump case
JMP	if (True)	Always Jump
JE / JZ	if (x == y)	Jump if Equal / Zero
JNE / JNZ	if (x != y)	Jump if Not Equal / Zero
JL (JLE)	if (x < y)	Jump if Less (or Equal)
JG (JGE)	if (x > y)	Jump if Greater (or Equal)

Process Flow Chart New



Topshiriq:

A=9, B=14 holatini tahlil qiling.

6-topshiriq. Flaglar asosida qaror qabul qilish

Quyidagi flaglar rolini izohlang:

- Zero Flag (ZF)
- Carry Flag (CF)
- Overflow Flag (OF)

Topshiriq:

Qaysi arifmetik amal qaysi flagni faollashtiradi? Misol bilan yozing.

7-topshiriq. Amaliy tekshiruv (Calculator bilan)

1. Calculator → Programmer mode
2. DEC va BIN rejimlarda
3. Ko'paytirish va bo'lishni bajaring

Xulosa yozing:

Assembler algoritmi bilan mos keldimi?

Hisobot tarkibi (talaba topshiradi)

1. Amaliy ish nomi: **A8 (2-qism)**
2. Maqsad
3. DX:AX tushunchasi
4. Murakkab ifoda hisoblash
5. Signed/Unsigned tahlil
6. Qoldiq va flaglar
7. Xulosa

Xulosa (namuna)

Ushbu amaliy ish davomida assembler tilida murakkab arifmetik amallarni chuqurlashtirilgan holda bajarish o'rganildi. Belgili va belgisiz hisoblashlar, DX:AX juftligi va flaglar asosida qaror qabul qilish mexanizmlari tahlil qilindi. Natijada assembler tilida real masalalarni algoritmik yechish ko'nikmasi shakllandi.

Nazorat savollari

1. DX:AX juftligi nima uchun ishlatiladi?
2. MUL va IMUL qachon tanlanadi?
3. Qoldiq qayerda saqlanadi?
4. Flaglar nima vazifa bajaradi?
5. Assemblerda murakkab ifodalar qanday bajariladi?